MILESTONE 2: RISK PORTFOLIO

Project: Portfolio Risk-Return Optimization

Roll No: IITMD_BA_2501070_FA-01

Date: February 15, 2026

1. Risk Metrics Analysis:

Using the 15 selected stocks from M1, we calculated comprehensive risk metrics:

1. **Volatility Analysis:**
   1. Annualized volatility was calculated for each stock using standard deviation of daily returns
   2. 14 stocks classified as "High Volatility" (>25%)
   3. 1 stock classified as "Medium Volatility" (HINDUNILVR at 23.39%)
   4. No low volatility stocks in the selection.

2. **Beta Calculation:**

All 15 stocks demonstrated defensive characteristics relative to Nifty 50 benchmark:

1. Beta range: 0.0138 to 0.0696
2. All stocks have Beta < 0.1 (highly defensive)
3. This indicates stocks moved less than the market during 2016-2020
4. Lowest Beta: HINDALCO (0.0138)
5. Highest Beta: BAJAJFINSV (0.0696)

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Volatility Analysis Complete

Stocks by Volatility:
        Symbol  Annualized_Return  Annualized_Volatility Volatility_Class
0   BAJAJFINSV            45.4306                36.7224             High
1        TITAN            43.5593                34.0834             High
2    NESTLEIND            30.5611                25.0529             High
3   ASIANPAINT            30.5339                25.6860             High
4     HINDALCO            34.8315                41.6577             High
5    KOTAKBANK            27.8717                28.3728             High
6   HINDUNILVR            26.6794                23.3891           Medium
7    TATASTEEL            29.6757                38.3201             High
8     SHREECEM            21.8721                30.2726             High
9         HDFC            20.8329                29.8451             High
10    JSWSTEEL            16.6894                54.4370             High
11        GAIL           -10.3308                43.9989             High
12      GRASIM            -6.2095                49.7712             High
13         IOC           -16.6222                46.2006             High
14   EICHERMOT            -4.7157                53.2020             High

Volatility Classification:
Volatility_Class
High      14
Medium     1
Low        0
Name: count, dtype: int64
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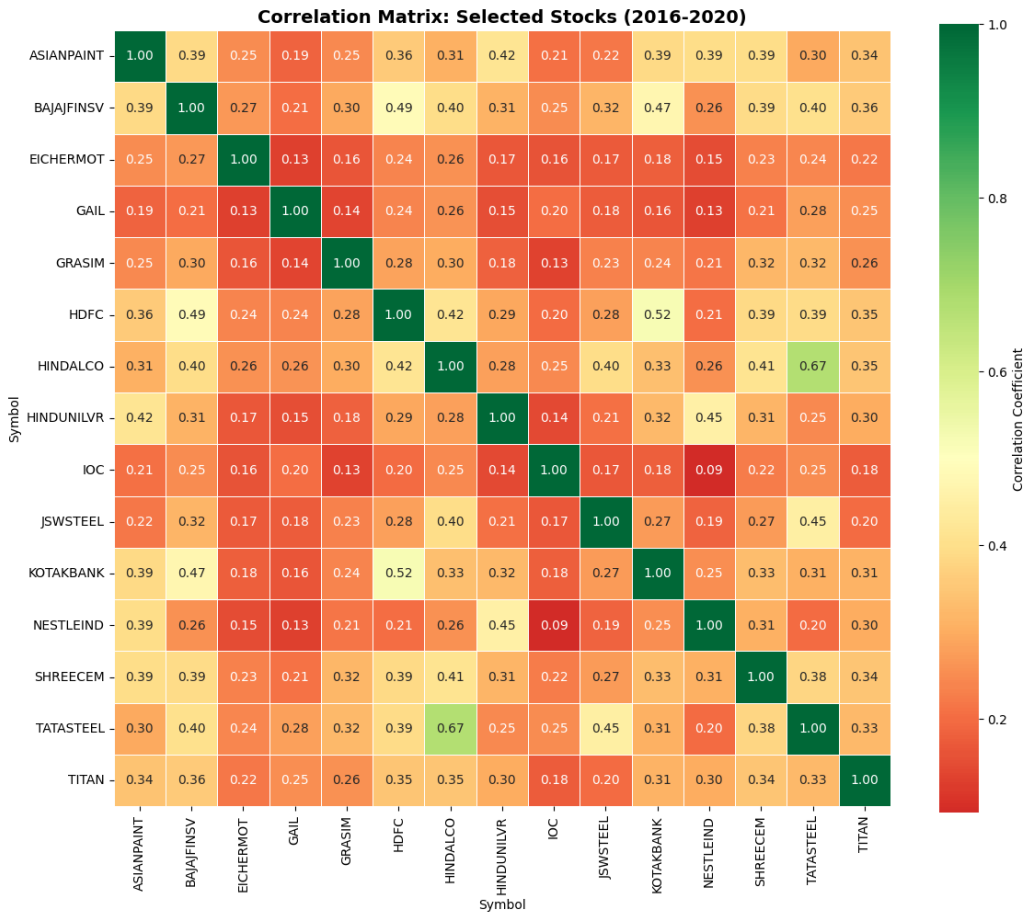
2. Correlation Analysis

A 15x15 correlation matrix was generated to assess diversification potential.

Key Findings:

- **Average correlation:** 0.30 (moderate)
- No stock pairs with correlation > 0.7 (indicating good diversification potential).
- **Highest correlation:** HINDALCO-TATASTEEL (0.67) - both metal sector stocks
- Many pairs with correlation < 0.3 (excellent for risk reduction).
- Low correlations enable effective portfolio diversification.



3. Portfolio Construction Methodology:

Three portfolios were constructed with different risk profiles using Beta-based classification:

Strategy:

Since all stocks have low Beta (<0.1), relative Beta rankings were used:

- Conservative: Stocks with lowest Beta (0.0138 - 0.0326)
- Balanced: Stocks across the Beta spectrum (0.0138 - 0.0696)
- Aggressive: Stocks with highest Beta (0.0465 - 0.0696)

All portfolios use equal-weight allocation (16.67% per stock).

A. Portfolio A: Conservative

- Target: Risk-averse investors, retirees, short-term goals
- Stocks: HINDALCO, GAIL, GRASIM, SHREECEM, IOC, ASIANPAINT
- Average Beta: 0.0216
- Strategy: Focus on most defensive stocks with lowest market sensitivity

B. Portfolio B: Balanced

- Target: Moderate risk appetite, medium-term goals
- Stocks: HINDALCO, SHREECEM, ASIANPAINT, TATASTEEL, JSWSTEEL, KOTAKBANK
- Average Beta: 0.0344
- Strategy: Mix of low, mid, and high Beta stocks for balanced exposure

C. Portfolio C: Aggressive

- Target: Risk-seeking investors, long-term goals, young investors
- Stocks: BAJAJFINSV, KOTAKBANK, TITAN, EICHERMOT, JSWSTEEL, HDFC
- Average Beta: 0.0521
- Strategy: Focus on stocks with highest Beta for maximum market exposure

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Comparison Table:
   Portfolio  Expected_Return  Risk_Volatility  Sharpe_Ratio  Average_Beta  Num_Stocks
Conservative           9.0125          24.1754        0.1039        0.0216           6
    Balanced          26.9124          25.3244        0.8060        0.0344           6
  Aggressive          24.9447          25.2828        0.7295        0.0521           6

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INTERPRETATION:
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Conservative Portfolio:
  - Expected Annual Return: 9.01%
  - Risk (Volatility): 24.18%
  - Sharpe Ratio: 0.1039
  - Average Beta: 0.0216

Balanced Portfolio:
  - Expected Annual Return: 26.91%
  - Risk (Volatility): 25.32%
  - Sharpe Ratio: 0.8060
  - Average Beta: 0.0344
...
  - Expected Annual Return: 24.94%
  - Risk (Volatility): 25.28%
  - Sharpe Ratio: 0.7295
  - Average Beta: 0.0521
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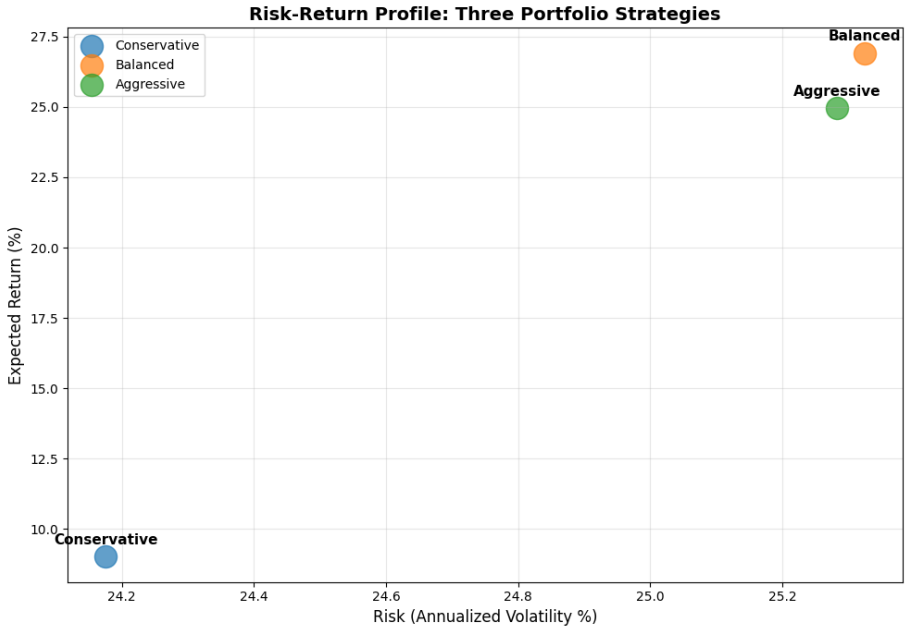
4. Portfolio Performance Comparison:

Portfolio metrics were calculated using Modern Portfolio Theory:

| Portfolio | Expected Return | Risk (Volatility) | Sharpe Ratio | Avg Beta |

⇒ **| Conservative | 9.01% | 24.18% | 0.10 | 0.0216 |**
⇒ **| Balanced | 26.91% | 25.32% | 0.81 | 0.0344 |**
⇒ **| Aggressive | 24.94% | 25.28% | 0.73 | 0.0521 |**

Risk-free rate assumed: 6.5% (10-year G-Sec)



Key Observations:

- Balanced portfolio offers best risk-adjusted returns (Sharpe: 0.81)
- Conservative portfolio has lowest return (9.01%) but also lowest risk
- Aggressive portfolio has high return (24.94%) with only marginally higher risk vs Balanced
- Minimal risk difference (1%) between Balanced and Aggressive portfolios

5. Conclusion & Recommendation:

Based on Sharpe Ratio analysis, the Balanced Portfolio is recommended for most investors as it provides:

- Highest risk-adjusted returns (Sharpe: 0.81)
- Optimal balance between return (26.91%) and risk (25.32%)
- Diversification across Beta spectrum.

The Conservative portfolio suits extremely risk-averse investors willing to accept lower returns, while the Aggressive portfolio is appropriate only for investors with high risk tolerance and long investment horizons.